MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11746284
POSITION-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Position-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746284	Circuit Card Assembly, Position-1

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- This assembly is one (1) of four (4) single ended printed circuit boards consisting of operational amplifiers and a sine/cosine module necessary to provide proper target positioning for the VIG.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746284.

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3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with the following power supply voltages:

a. +15.0 VDC
b. -15.0 VDC
c. 0.000 to +10.000 VDC
b. 0.000 to -10.000 VDC

3.2.1.2 Integrating Network #1.- The operational amplifier network shall be capable of integrating inputs and yielding the outputs as shown in Table 1.

3.2.1.3 Integrating Network #2.- The operational amplifier network shall be capable of integrating input and yielding the output as shown in Table 2.

3.2.1.4 Integrating Network #3.- The operational amplifier network shall be capable of integrating inputs and yeilding the outputs as shown in Table 3.

3.2.2 Environmental.- The assembly shall meet the environmental requirements specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11746284 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746284 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

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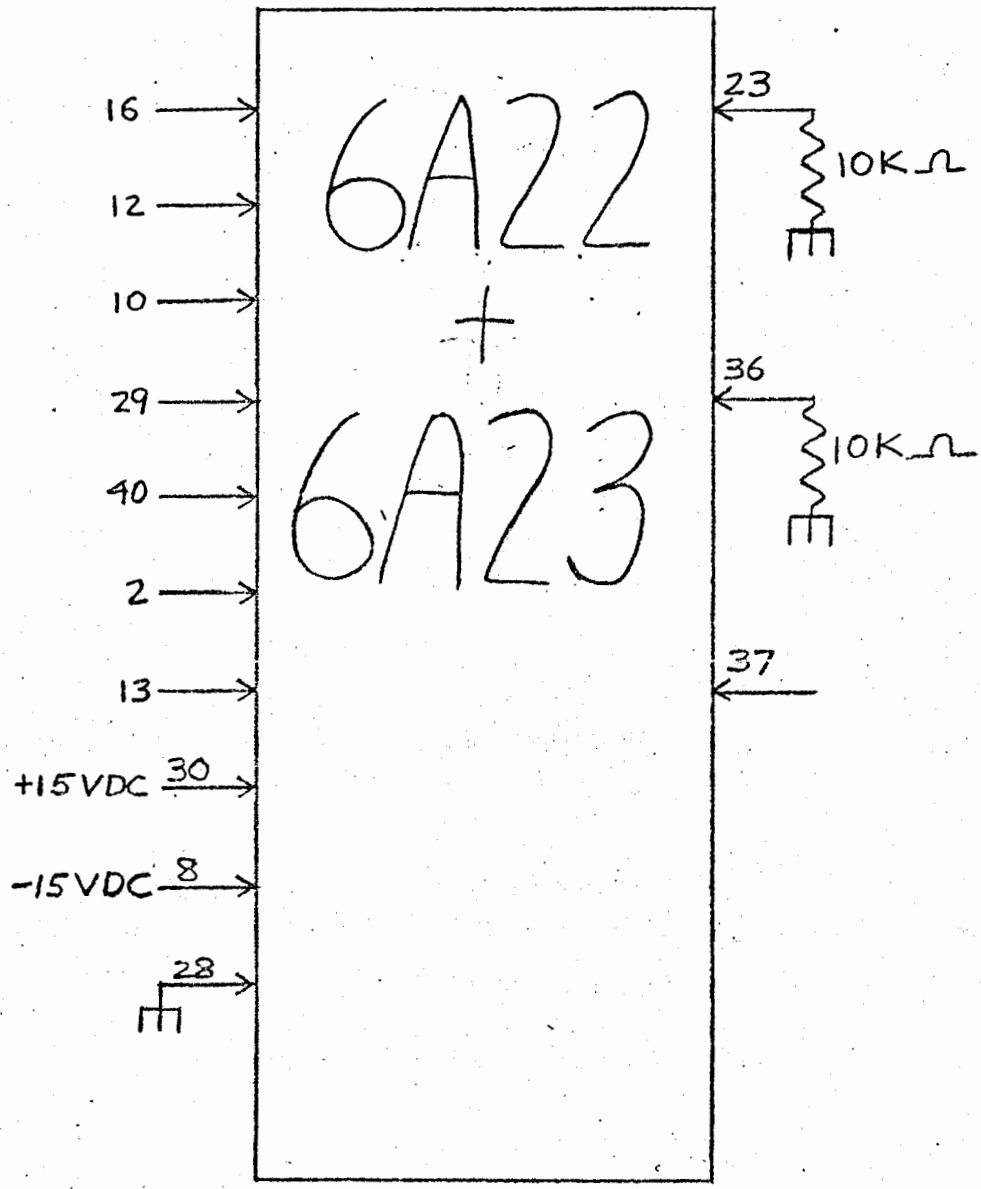


FIGURE I

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4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and counter testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment. (see Figure 1)

4.2.1.1 Power Supply.- The power supplies shall be as follows:

a. +20.0 VDC ± 0.1 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

b. -20.0 VDC ± 0.1 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

c. (0.000 to +15.000) VDC ± mV DC, 100 mA, or greater and a ripple of less than 10 V RMS.

d. (0.000 to -15.000) VDC ± 1 mV DC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Dummy Loads.- The dummy loads shall be two 10K ohm resistors.

4.2.1.3 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input impedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these test shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power Supply.- All requirements shall be met with the following power supplies:

a. +15.0 VDC ± 0.1 VDC
b. -15.0 VDC ± 0.1 VDC
c. (0.000 to +10.000) VDC ± 1 mV VDC
d. (0.000 to -10.000) VDC ± 1 mV VDC

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4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-Up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.3 Performance.- When voltages of between ± 10 VDC are applied to the integrator network input, the corresponding output shall be as shown in Tables I, II and III.

TABLE I

INPUT (VDC)			OUTPUT (VDC)
Pin 16	Pin 12	Pin 10	Pin 23
1V ± 1 mV	0V ± 1 mV	0V ± 1 mV	6.750 ± .200
0V ± 1 mV	-10V ± 1 mV	0V ± 1 mV	-1.700 ± .200
0V ± 1 mV	0V ± 1 mV	2V ± 1 mV	9.000 ± .200

TABLE II

INPUT (VDC)			OUTPUT (VDC)
Pin 29	Pin 40	Pin 2	Pin 36
1V ± 1 mV	0V ± 1 mV	+2V ± 1 mV	3.684 ± .200
0V ± 1 mV	-1V ± 1 mV	-2V ± 1 mV	-3.680 ± .200
0V ± 1 mV	0V ± 1 mV	+1V ± 1 mV	-6.040 ± .200

TABLE III

INPUT (VDC)	INSTRUCTION	OUTPUT (VDC)
Pin 13		Pin 37
0V ± 1 mV	Adjust R14 to obtain	0V ± 1 mV
1V ± 1 mV		.638 ± .100
3V ± 1 mV		1.940 ± .100
5V ± 1 mV		3.330 ± .100
7V ± 1 mV		4.936 ± .100
9V ± 1 mV		7.129 ± .100

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NOTE: The output tolerances during testing at temperature extremes shall be ± 0.300 VDC.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746284.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A613

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